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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

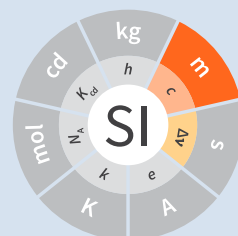
**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

Part 2

Iodine ()

Iodine ()



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Iodine ()**

CCL-CCTF Frequency Standards Working Group

9th edition 2003

1. CIPM recommended values

The values $f = 551\,580\,162\,400\text{ kHz}$
 $\lambda = 543\,515\,663.608\text{ fm}$

with a relative standard uncertainty of 4.5×10^{-11} apply to the radiation of a laser stabilized to an external iodine cell and subject to the following conditions:

- cold point temperature $(0 \pm 2)^\circ\text{C}^{(1)}$;
- frequency modulation width, peak-to-peak, $(2.0 \pm 0.5)\text{ MHz}$.

⁽¹⁾ For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols $\pm$ refer to a tolerance, not an uncertainty.

2. Source data

Adopted value $f = 551\,580\,162\,400\,(25)\,\text{kHz}$
 $u_c/y = 4.5 \times 10^{-11}$
 for which:
 $\lambda = 543\,515\,663.608\,(24)\,\text{fm}$
 $u_c/y = 4.5 \times 10^{-11}$

calculated from

f/kHz	u_c/y	source data
551 580 162 397.1	1.3×10^{-11}	[1]

Given the small number of calibrations and the individual behaviour of participating lasers in this calibration, the CCL considered it prudent to adopt an uncertainty of 25 kHz for the recommended value. The final CCL value chosen was $f = 551\,580\,162\,400\,(25)\,\text{kHz}$ with the following operating conditions:

- cold point temperature $(0 \pm 2)^\circ\text{C}^{(1)}$;
- frequency modulation width, peak-to-peak, $(2.0 \pm 0.5)\,\text{MHz}$.

It was felt appropriate not to define the laser type.

⁽¹⁾ For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols $\pm$ refer to a tolerance, not an uncertainty.

3. Absolute frequency of the other transitions related to those adopted as recommended and frequency intervals between transitions and hyperfine components

These tables replace those published in BIPM Com. Cons. Long., 2001, **10**, 182-183 and *Metrologia*, 2003, **40**, 126.

The notation for the transitions and the components is that used in the source references. The values adopted for the frequency intervals are the weighted means of the values given in the references.

For the uncertainties, account has been taken of:

- the uncertainties given by the authors;
- the spread in the different determinations of a single component;
- the effect of any perturbing components;
- the difference between the calculated and the measured values.

In the tables, u_c represents the estimated combined standard uncertainty (1σ).

All transitions in molecular iodine refer to the B-X system.

Table 1

$\lambda \approx 543 \text{ nm } ^{127}\text{I}_2 \text{ R(12) 26-0}$					
a_n	$[f(a_n) - f(b_{10})]/\text{MHz}$	u_c/MHz	a_n	$[f(a_n) - f(b_{10})]/\text{MHz}$	u_c/MHz
a_1	-1 162.24	0.02	a_9	-679.420	0.005
a_2	-909.87	0.02	a_{10}	-596.134	0.005
a_3	-900.11	0.03	a_{11}	-485.61	0.01
a_4	-853.336	0.005	a_{12}	-476.35	0.01
a_5	-848.131	0.005	a_{13}	-423.23	0.01
a_6	-795.92	0.01	a_{14}	-410.01	0.01
a_7	-752.382	0.005	a_{15}	-305.910	0.005
a_8	-733.134	0.005			
Frequency b_{10} , R(106) 28-0, $^{127}\text{I}_2$: $f = 551\,580\,162\,400 \text{ kHz}$ [2] referenced to					

Ref. [3], [4], [5], [6], [7], [8], [9]

Table 2

$\lambda \approx 543 \text{ nm } ^{127}\text{I}_2 \text{ R(106) 28-0}$					
b_n	$[f(b_n) - f(b_{10})]/\text{MHz}$	u_c/MHz	b_n	$[f(b_n) - f(b_{10})]/\text{MHz}$	u_c/MHz
b_1	-573.765	0.005	b_9	-114.575	0.005

b ₂	-320.462	0.005	b ₁₀	0	—
b ₃	-291.59	0.01	b ₁₁	124.83	0.01
b ₄	-282.143	0.005	b ₁₂	132.31	0.01
b ₅	-253.675	0.005	b ₁₃	154.51	0.01
b ₆	-172.693	0.005	b ₁₄	162.65	0.01
b ₇	-159.428	0.005	b ₁₅	287.24	0.01
b ₈	-127.760	0.005			
Frequency	b ₁₀ , R(106) 28-0, ¹²⁷ I ₂ : <i>f</i> = 551 580 162 400 kHz [2]				
referenced to					

Ref. [3], [4], [5], [6], [7], [8], [9]

Absorbing molecule $^{127}\text{I}_2$, b_{10} component, R(106) 28-0 transition⁽¹⁾

⁽¹⁾ All transitions in I_2 refer to the $\text{B}^3\Pi\ 0_u^+ - \text{X}^1\Sigma_g^+$ system.

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